

CS101

Boolean Logic

2/12

Agenda

1. Quiz4
2. HW2 Demo and Qs
3. Alex Lee: Logical Operators
4. Order of Operations Worksheet
5. Lab5 Work time
6. Windows OS Tips + Touch Typing

Announcements

1. Lab1 and Lab2 due tomorrow 2/13 at midnight
2. No Class 2/17
3. No Quiz 2/19
4. Lab3 and HW2 Due Friday 2/20

HW2 Demo

Logical Operators In Java #20

<https://www.youtube.com/watch?v=p0UCTkqU2mg>

Java Logical Operators

<i>op</i>	<i>meaning</i>	<i>true</i>	<i>false</i>
&&	<i>AND (both true)</i>	<code>true && true</code>	<code>true && false</code>
 	<i>OR (at least one true)</i>	<code>true false</code>	<code>false false</code>
!	<i>NOT (inverts boolean)</i>	<code>!false</code>	<code>!true</code>

Truth Table for Logical Operators

<i>A</i>	<i>B</i>	<i>A && B</i>	<i>A B</i>
true	true	true	true
true	false	false	true
false	true	false	true
false	false	false	false

Java Comparison Operators

<i>op</i>	<i>meaning</i>	<i>true</i>	<i>false</i>
==	<i>equal</i>	2 == 2	2 == 3
!=	<i>not equal</i>	3 != 2	2 != 2
<	<i>less than</i>	2 < 13	2 < 2
<=	<i>less than or equal</i>	2 <= 2	3 <= 2
>	<i>greater than</i>	13 > 2	2 > 13
>=	<i>greater than or equal</i>	3 >= 2	2 >= 3

Order of Operations

Common operators, from highest to lowest priority

HIGHEST	()	Parentheses
	**	Exponent
	* / %	Multiplication, Division, Modulus
	+ -	Addition, Subtraction
	> < >= <=	Comparison
	== !=	Equality
	&&	Logical AND
		Logical OR
LOWEST	= += -= *= /=	Assignment

Order of Operations Worksheet

Lab5 Work time

Before Next Class:

- Submit Labs!
- Lab3 and HW2 due Friday Feb 20th

Next Class 2/19

1. Compiled vs Interpreted
 2. Continue Booleans+Conditionals
- **Bring Headphones every class!**